Computational references are not experiments: pre-registered validation of machine-learned sodium-cathode voltages

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Machine-learning screens for battery materials are trained and judged almost entirely against computed reference voltages, and those references carry their own systematic errors. We report a case in which this matters quantitatively: our own screening stack (a graph-network voltage screen, a prior-art triage layer, and a local PBE+U bench) fails pre-registered validation against experiment-anchored literature values. Verdict thresholds, failure modes, and the primary metric were committed before analysis. On an operator-audited set of known Na-ion cathodes ($n = 6$ after one documented exclusion; verdict unchanged at $n = 7$), the raw held-out mean absolute error was 0.67 V, the pre-registered conservative metric, the upper 95% confidence bound of the cross-validated bias-corrected error, was 1.09 V, and the residual was strongly voltage-dependent ($r = -0.94$), so no additive calibration is valid. On the two compounds where prediction, database reference, and experiment could all be compared, the Materials Project PBE+U reference sat about 0.54 V below measurement: the reference, not the model, dominated the error. A prior-art screen found at least 70% of the targeted Na substitution space already published. We retire the screen, bound what "verified" means for our DFT ledger, and pre-register a calibration audit of it against four benchmark Li couples.

I. INTRODUCTION

Generative models now propose inorganic crystals far faster than any laboratory, or any density-functional-theory (DFT) queue, can check them. The GNoME release alone reported 2.2 million candidate structures and declared roughly 380,000 of them stable [1], and an autonomous laboratory reported synthesizing dozens of those candidates within weeks [2]. The published record since then has been less kind. Cheetham and Seshadri found that almost none of the examined GNoME compounds were simultaneously credible, new, and useful by the standards a chemist would apply [3], and a re-examination of the autonomous-synthesis claims concluded that none of them was convincingly demonstrated [4]. The field's binding constraint has moved from generating candidates to deciding which claims about them deserve trust.

That decision rests on reference data more often than the literature acknowledges. Screening models for battery cathodes are typically trained on intercalation voltages computed within GGA or GGA+U [5, 6], most commonly the Materials Project values [7], and they are usually *evaluated* against the same computed quantities. A model can therefore pass every benchmark available to it while inheriting, invisibly, whatever systematic error separates its reference scale from the electrochemical measurements the field actually cares about. Validation against experiment is the only test that can expose this, and it is the test least often run.

This paper runs that test on our own system, under rules fixed in advance. The system under test (Sec. II) is a small closed-loop materials pipeline (generator, graph-network voltage screen, prior-art triage layer, and a local PBE+U verification bench) built and operated by one person. Before any validation metric was computed, we committed verdict thresholds, a conservative primary metric, and four named failure modes to the repository, and a fifth stop condition was fixed before literature curation began, so that the result could not be renegotiated after the fact (Sec. III); the commit hashes are the audit trail [8]. The outcome is negative on every axis we pre-registered. The voltage screen is not screening-grade against experiment, and its residual structure forbids the additive recalibration that would otherwise be the obvious patch (Sec. IV). The computed references themselves sit about half a volt below measurement on the rows where a three-way comparison was possible, which redistributes blame in an uncomfortable direction: part of what the model learned to reproduce was reference error, not electrochemistry (Sec. V). And the targeted "discovery" space the generator was aimed at was at least 70% already published, before any question of model accuracy arises. The constructive output is the protocol itself, a small quote-anchored experimental reference set with documented provenance, and a pre-registered calibration audit of our own DFT scale whose decision rule is committed but whose verdict, at the time of writing, does not yet exist (Sec. VI).

We claim no discovered material, no validated screen, and no calibrated scale. The paper's contribution is the discipline: every quantitative claim below traces to a committed artifact, every failure is reported under the definition fixed before the analysis ran, and the strongest objections we know of are stated and answered in Sec. VII.

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II. SYSTEM UNDER TEST

The pipeline, called QME, is a solo-operated closed loop that runs on a single Apple M4 desktop. One cycle proceeds: generate candidate crystals, screen them with a property-predicting graph neural network (GNN), rank survivors with a Gaussian-process acquisition function over learned embeddings, screen the ranked list for prior art, verify the top picks with paired DFT calculations, and fine-tune the GNN on the newly verified results. Each stage is described here at the level needed to interpret the validation; computational details are in Sec. VIII.

Candidates come from prototype substitution: redox-site substitution into known framework structures retrieved from the Materials Project (MP) [7], followed by charge-neutrality and electronegativity screening with SMACT [9]. The screen is a Siamese multi-head GNN that ingests a (charged, discharged) structure pair and predicts the average intercalation voltage with a Monte-Carlo-dropout uncertainty. Its training corpus contains 2814 graphs, of which 20 pairs (0.7%) are Na chemistries; the model is, by construction, a Li-dominated predictor asked to extrapolate. A second network, used for stability estimates and for the embedding space, is not ours: it is the universal interatomic-potential backbone MACE-MP-0 [10, 11] with property heads, and we treat it throughout as borrowed machinery. The acquisition function is a plain upper-confidence-bound rule over a PCA reduction of those embeddings; a diversity-weighted variant exists in the codebase but was never wired into the orchestrator, and with five verified training points the acquisition stage has no demonstrated value of any kind. We make no claim for it.

The prior-art layer is the part of the stack built to say *no*. A candidate is flagged when a literature, structure-database, or family-pattern source names its composition: citations count only if a chemical formula in the retrieved record matches the candidate's composition exactly, as a de-intercalated framework, or as the same element set, which removes the loose-token false positives that plague relevance-ranked search. The layer is safety-locked: it can downgrade a candidate to known but can never certify novelty, and its machine verdict for absence is bounded by the coverage of the sources it actually searched. In its committed precision audit it flagged 8 of 8 known compounds and false-flagged 0 of 5 fictional control compositions.

Verification is paired-state DFT: both members of each (charged, discharged) pair are relaxed under identical PBE+U settings with Quantum ESPRESSO [12, 13], and the average voltage follows from the total-energy difference (Sec. VIII). The verified ledger contains seven rows: five known Li-ion cathodes that serve as calibration anchors ($LiCoO_2$ at 4.120 V, $LiFe(PO_3)_4$ at 5.687 V, $LiNiP_2O_7$ at 5.232 V, $LiFeP_2O_7$ at 5.275 V, $Li_2Fe(PO_3)_5$ at 5.6115 V), one experimental touchpoint, and one computed-only couple with no experimentally anchorable counterpart (Appendix D). Every row is voltage-only: hull distances were not computed and kinetic screens were never run, so "verified" in this paper always means a converged voltage on a consistent PBE+U scale and nothing more. Four of the five anchors are metaphosphate or pyrophosphate couples between 5.2 V and 5.7 V, outside any practical electrolyte window and, to our knowledge, without experimental electrochemistry at those couples. The ledger's single experimental touchpoint is $Na_3V_2(PO_4)_3$, computed at 2.8957 V against a measured plateau near 3.4 V [14]. Zero candidates classified as absent from our corpora have ever been verified, so nothing in the ledger tests discovery.

The loop has closed once: the fifth anchor triggered a fine-tune of the GNN, gated on a held-out leave-one-out cross-validation over the five anchors and on the fitted anchor error (0.084 V mean absolute), and the fine-tuned model was activated and used for every prediction in this paper. Both gates compare the model to its own DFT anchors. Whether that self-referential standard survives contact with experiment is the question the rest of the paper answers.

III. PRE-REGISTERED PROTOCOL

The validation question was fixed as: can the active GNN predict Na-ion average voltages well enough to drive screening decisions? Before computing any bias-corrected metric we committed a verdict ladder, a primary metric, and four failure modes (F1–F4) to the repository; a fifth stop condition (F5) was fixed in the Step-3 protocol before literature curation began. The committed documents, their hashes, and the verbatim definitions are reproduced in Appendix C.

The ladder maps the held-out mean absolute error (MAE) to one of three verdicts: screening-grade below 0.30 V, ranking-only between 0.30 V and 0.50 V, and not screening-grade above 0.50 V. The primary metric is deliberately conservative: the upper edge of the 95% bootstrap confidence interval on the held-out, bias-corrected MAE, where "held-out" requires that no compound's error contributed to the bias estimate applied to it. Point estimates, in-sample numbers, and best-family numbers are reported but cannot set the verdict.

The failure modes encode the ways a calibration claim could be technically true and practically meaningless. F1 fires when per-family bias estimates spread by more than 0.15 V, in which case a single additive correction does not generalize and no clean verdict is issued. F2 fires on any in-sample contamination of the bias estimate. F3 marks any verdict from fewer than twenty held-out compounds as provisional. F4 fires when family biases disagree in sign. F5 is a curation stop: if more than half of the compounds in the experimental upgrade cannot be cited from primary literature, the subset is not defensibly experimental, and the analysis halts rather than estimating its way to a sample. The pre-registration also commits revision triggers that outlast publication: among them, two in-box compounds disagreeing with corrected predictions by more than 0.4 V retire the bias-corrected predictor entirely.

Reference curation followed a quote-anchored discipline. Every literature voltage carries a verbatim snippet from a source fetched during the run, a DOI, the cell configuration, and the reference electrode; values against anything other than Na metal were rejected. Each datum was assigned an evidence grade, defined at curation time and not pre-registered, of A (average voltage stated by the authors in fetched text), B (single arithmetic step from author-stated plateaus or phase windows, calculation recorded), or C (read off a published figure; low confidence). Grade D (no anchorable source) meant the compound was dropped, and five compounds were dropped rather than estimated (Appendix B). Every polymorph was lattice-verified against the MP structure the GNN actually ingested, because the screen predicts on a specific crystal, not on a formula. Finally, the curated table went to a human audit: the operator manually verified the five most verdict-sensitive extractions against the primary sources, all five of which were confirmed, and that audit reclassified one row (maricite $NaFePO_4$) as a phase-identity mismatch, which defines the canonical $n = 6$ metric set used below.

IV. RESULTS A: THE SCREEN AGAINST REFERENCES, THEN AGAINST EXPERIMENT

A. Against computed references

The held-out test that motivated everything else used 74 Na-ion cathode entries from the MP battery database whose discharged structures do not appear in the training corpus, with the MP computed average voltage as the reference. The raw held-out MAE was 0.5892 V with a mean signed error of 0.4263 V and a Pearson correlation between prediction and reference of 0.28. Most of the error, in other words, looked like a single large positive bias, which is exactly the situation an additive correction is supposed to repair, and exactly where the pre-registered machinery earned its keep.

The bias is not one number. Grouped by chemistry family (Table I), the per-family signed errors span 0.4419 V between the mixed-anion "other" bucket (0.6316 V, $n = 38$) and the layered oxides (0.1897 V, $n = 9$), with the polyanionic phosphates (the family the Na campaign targeted) at 0.2022 V ($n = 26$). That spread exceeds the pre-registered 0.15 V threshold, so F1 fired and no clean verdict tier could be issued. The leave-one-family-out estimate of what an additive correction would actually deliver made the point mechanically: the corrected held-out MAE ($n = 64$) was 0.5644 V, *worse-behaved* than it appears since the residual mean bias of 0.1101 V should have been near zero if the correction generalized, and the conservative upper 95%-confidence value was 0.6610 V. Cross-family bias transfer moves error around instead of removing it.

TABLE I. Per-family raw bias and MAE of the active GNN against MP computed reference voltages on the 74-compound held-out set. The spread between family biases (families with $n \geq 5$) is 0.4419 V, which fires pre-registered failure mode F1: no single additive correction generalizes across families.

Family	n	bias (V)	MAE (V)
polyanionic phosphate	26	+0.2022	0.3753
layered oxide	9	+0.1897	0.4236
other (mixed-anion, fluorides)[a]	38	+0.6316	0.7750
NASICON	1	+0.5834	0.5834

[a] Mostly carbonate-polyanionic (P/As/Si–C–O) and halide chemistries, the least represented in the training corpus.

These numbers compare the model to MP's computed voltages, so they bound the model's agreement with its own reference scale, not with electrochemistry. The pre-registered plan was therefore to upgrade the reference on the phosphate subset to literature experimental values. That upgrade failed in a way we did not anticipate and now consider a finding in its own right: of the 26 held-out polyanionic phosphates, the number with a citable experimental average voltage in primary literature was zero. Eighteen of the 26 have no Na-cell publication of any kind; the remainder appear only in structural crystallography or report cycling windows without an extractable average. The stop condition F5 (threshold: half the subset uncitable) fired at 100%, and the analysis halted rather than substituting estimates. The held-out set, drawn from a computed battery database, was not a set of materials anyone had measured. A reference upgrade therefore required building a new compound set from the experimental literature inward, which is Step 3b.

B. Against experiment

The curated set contains nine literature-validated rows: five evidence grade A, three grade B, one grade C, spanning phosphates, fluorophosphates, one sulfate, and one layered oxide (Appendix B, Table V). Two rows ($Na_2FeP_2O_7$ and alluaudite $Na_2Fe_2(SO_4)_3$) have no MP structure for the GNN to ingest and are excluded from metrics; seven were predicted. The operator audit then excluded maricite $NaFePO_4$ for phase identity: its 2.6 V literature value belongs to the amorphous phase formed on first desodiation [15], while the model predicted on crystalline maricite, leaving $n = 6$ as the canonical set. Every verdict below is unchanged at $n = 7$.

The raw held-out MAE on the canonical set is 0.668 V with a mean signed error of 0.231 V (Table II). Applying the pre-registered leave-one-out additive bias correction makes the error larger, not smaller: 0.802 V corrected, with an upper 95% confidence edge of 1.092 V, which is

TABLE II. Held-out error of the active GNN against experimental average voltages, for all computed variants. The pre-registered primary metric is the upper edge of the 95% bootstrap confidence interval (CI) on the bias-corrected MAE; every variant of every estimate, including every lower CI edge of the corrected metric, exceeds the 0.50 V "not screening-grade" threshold.

Set	n	raw MAE	raw bias	raw CI_{95}	corr. MAE	corr. CI_{95}
held-out, all grades	7	0.694	+0.319	0.959	0.756	1.017
held-out, A/B only[a]	6	0.668	+0.231	0.977	0.802	1.092
phosphate, all grades	4	0.654	+0.045	0.816	0.872	1.088
phosphate, A/B only	3	0.590	−0.222	0.784	0.774	1.161

[a] The canonical set after the operator-audit exclusion. CI_{95} columns give the upper edge of the 95% bootstrap interval on the MAE; the corrected upper edge of this row is the pre-registered primary metric. All values in volts.

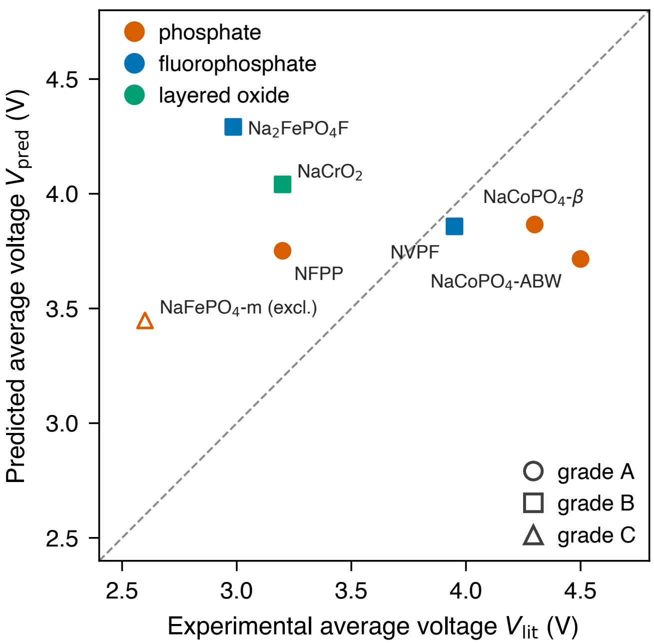

FIG. 1. Active-model predictions against quote-anchored experimental average voltages for the curated Na-ion cathode set, colored by chemistry family with evidence grades marked. The open symbol is maricite $NaFePO_4$, excluded from the canonical metrics after the operator audit identified a phase-identity mismatch (the cycled phase is amorphous, the predicted structure crystalline). The diagonal is parity. Predictions compress into a band near 3.4–4.3 V across references spanning 2.6–4.5 V.

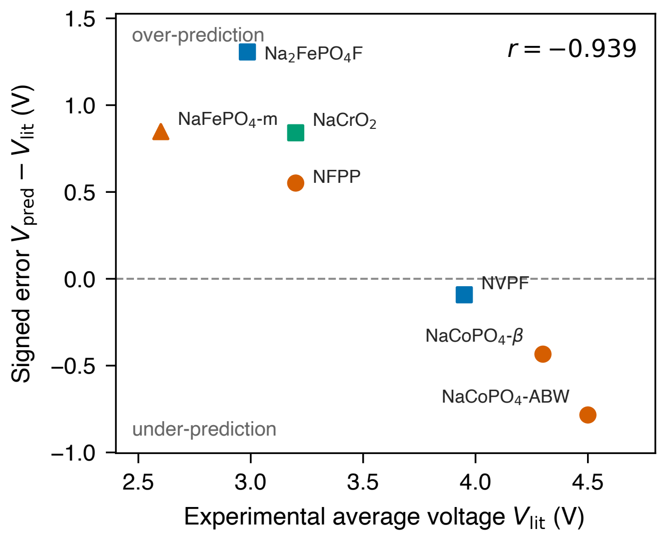

FIG. 2. Signed prediction error against the experimental reference voltage for the seven predicted rows. The correlation $r = -0.939$ quantifies the compression: over-prediction below ∼3.5 V, under-prediction of the 4.3–4.5 V cobalt rows. No additive correction can remove a residual with this structure, which is why the pre-registered correction worsens the MAE (Table II).

the pre-registered primary metric. The mechanism is visible in Fig. 2: the signed error is almost a linear function of the reference voltage, with Pearson $r(\mathrm{err}, V_{\mathrm{lit}}) = -0.939$. The model over-predicts every compound below about 3.5 V (maricite 0.85 V, NFPP 0.55 V, Na_2FePO_4F 1.31 V, $NaCrO_2$ 0.84 V) and under-predicts the two high-voltage cobalt polymorphs (−0.78 V and −0.43 V), compressing a 1.9 V experimental range into roughly half that span of predictions. A single additive shift cannot repair a sign-flipping, voltage-dependent residual; subtracting the mean bias improves the low-voltage rows while pushing the cobalt rows further from the measurements. This is the experimental confirmation of the F1 structure seen against computed references in Sec. IV A.

Against the pre-registered ladder, the conservative primary metric of 1.092 V sits far above the 0.50 V threshold: the screen is *not screening-grade*, with the verdict marked provisional because F3 fired ($n < 20$). The verdict needs no benefit of the doubt: the raw point estimate, the corrected point estimate, and both confidence edges of every computed variant (with and without the grade-C row; full set and phosphate subset) all individually exceed 0.50 V (Table II). The pre-registered revision trigger also engaged: two in-box iron phosphates miss by more than the 0.4 V trigger threshold in raw error, which retires the bias-corrected predictor outright. As a reproducibility check, the two rows shared with the Sec. IV A analysis reproduce their earlier predictions within Monte-Carlo-dropout noise ($\Delta = -0.011$ V and -0.055 V). The operator audit confirmed all five spot-checked extractions, so

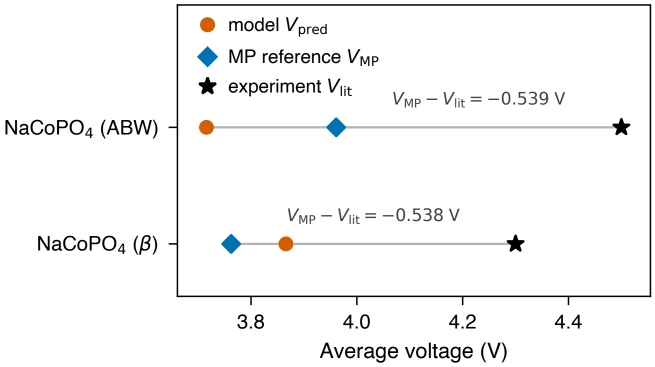

FIG. 3. Three-way comparison for the two polymorph-resolved $NaCoPO_4$ rows: model prediction, MP computed reference, and quote-anchored experimental value. The reference sits ≈0.54 V below measurement on both polymorphs; the model–reference difference is smaller and changes sign.

the failure cannot be attributed to careless curation. The screen, evaluated exactly as we committed to evaluate it, does not work.

V. RESULTS B: THE REFERENCE SCALE AND THE PRIOR-ART BASE RATE

A. Decomposing prediction error against the reference

Two curated rows, the ABW and β polymorphs of $NaCoPO_4$, also appear in the MP battery database with computed average voltages, which permits the three-way decomposition $(V_{pred} - V_{lit}) = (V_{pred} - V_{MP}) + (V_{MP} - V_{lit})$ on polymorph-resolved structures. The result is lopsided (Fig. 3). The MP PBE+U reference sits 0.539 V below the measured value for the ABW polymorph and 0.538 V below it for β, the same offset twice, while the model's deviation from its own reference is smaller and mixed in sign (-0.245 and 0.104 V). For the β polymorph the model lands *closer* to experiment than the reference it was trained toward. On these rows, most of the prediction-versus-experiment gap is reference error, not model error.

Two caveats bound this finding, and we state them at full strength. First, $n = 2$: this is a direction, not an estimate. Second, the $NaCoPO_4$ literature values are first-charge averages from cells with roughly 11% first-cycle reversibility [16], and kinetic polarization inflates a first-charge average above the equilibrium voltage, so part of the 0.54 V is experimental upper bias rather than DFT error. What makes the direction credible despite $n = 2$ is independent replication on our own bench: our PBE+U calculation of $Na_3V_2(PO_4)_3$ (a different compound, a different code installation, an empirically fitted U_V) gives 2.8957 V against a measured plateau near 3.4 V [14], an offset of the same sign and similar size. Published GGA+U benchmarking reports underestimation of transition-metal redox voltages of comparable scale when U is not tuned per couple [5, 6]. Three independent comparisons agree in direction; none alone is conclusive.

The consequence reaches beyond our pipeline. The Sec. IV A family analysis found the model predicting 0.20 V *above* MP references on phosphates, which looked like model bias; against experiment, those same predictions are too low. A model trained toward MP voltages in this chemistry inherits a deficit of roughly 0.4–0.5 V against measurement, and the better it fits its references the more faithfully it reproduces their offset. Any screen whose accuracy statement ends at "versus MP" should be read with that substitution in mind.

B. Reference integrity as a finding

Building the curated set surfaced a class of errors that no accuracy metric captures: identity errors in the reference chain. Machine validation of the initial human-drafted curation proposal found six of its MP identifiers wrong. One was a namespace collision: the battery-database identifier matching $Na_4Fe_3(PO_4)_2(P_2O_7)$ actually keys a Cr_3O_8 battery pair, so a training-set membership flag based on identifier matching was simply false. The others pointed at unrelated compounds ($Fe_6O_5F_7$, $CaWO_4$, $Li_3MnCr_3O_8$, $Na_7(CoO_3)_2$) or did not exist. Phase identity was the other recurring trap. The maricite $NaFePO_4$ literature voltage belongs to an amorphous phase created by the first charge [15]; the MP entry alleged to be maricite $NaMnPO_4$ is a different compound ($Na_2Mn_3(PO_4)_3$); and the literature identity of $NaVPO_4F$ is itself disputed between a tavorite phase and $Na_3V_2(PO_4)_2F_3$ [17]. Each of these, uncaught, silently corrupts a validation pair: the model predicts on one crystal while the reference describes another. The full correction and drop log is Appendix B. We draw a methodological conclusion: in this regime, reference-data integrity, not model architecture, is the binding constraint on what a validation can even mean.

C. Base rates in the targeted space

A screen also cannot be useful if the space it screens contains nothing left to find. Enumerating the generator's own targeted space (earth-abundant Na cathode compositions over its standard framework library) gives 390 distinct formulas, of which 78 survive charge-neutrality and electronegativity screening. The prior-art layer, which can only downgrade, flags 55 of those 78 (70.5%) as already published: 52 by structural-family match, 20 by exact composition in the Crystallography Open Database, 2 by composition-verified literature citation (paths overlap). Because the layer certifies nothing and two of its sources are stubs, 70.5% is a lower bound. The 23 surviving formulas are substitutional variants of two known families, and at least one survivor is a known

TABLE III. Prior-art base-rate probes. Top: the generator's targeted earth-abundant Na space. Bottom: a pinned uniform sample of the GNoME stable-materials release screened with the same layer, and the staged triage funnel applied to the prior-art-absent remainder. Flagged rates are lower bounds (the layer only downgrades); the GNoME verified floor reflects a hand audit of the five flagged hits (3/5 confirmed). The stability disputes are a disagreement rate between two models, not a correctness measure.

Quantity	Value
Targeted earth-abundant Na space (substitution generator)	
enumerated compositions	390
SMACT-valid	78
flagged already-published (lower bound)[a]	55 (70.5%)
via structural family	52
via COD exact composition	20
via named literature citation	2
survivors (absent within coverage)	23
GNoME stable-materials release (pinned uniform sample)	
population / sample	554,054 / 500
flagged by title-level screen	5 (1.0%)
hand-verified true positives[b]	3/5 (floor 0.6%)
COD exact-composition matches	0/500
prior-art-absent carried to funnel	495
after SMACT validity	49
after stability screen	36
after polaron screen (survivors)	35 (7.1%)
stability disputes vs GNoME label[c]	31.1%

[a] The prior-art layer only downgrades and two of its sources are stubs; flagged rates are lower bounds. Paths overlap.
[b] Hand audit of every flagged hit.
[c] Disagreement rate between the borrowed stability model and the GNoME label; a correctness claim for neither.

cathode the title-level checker missed, a measured precision disclosure rather than a footnote. The targeted space is saturated in exactly the sense Cheetham and Seshadri describe for generated libraries at large [3].

To test whether that base-rate problem is ours alone, we ran the same layer over a pinned uniform random sample ($N = 500$, fixed seed and identifier list) of the 554,054-entry GNoME stable-materials release [1], followed by our staged triage funnel on the 495 prior-art-absent survivors (Table III). The title-level screen flagged 1.0% of the sample; hand-auditing those five hits confirmed three, a verified floor of 0.6%, and an exact-composition check against the Crystallography Open Database matched 0 of 500. Read correctly, these numbers say the checkable-known fraction of GNoME under *this* layer's coverage is small: they measure our coverage as much as GNoME's novelty, and they are consistent with the published finding that GNoME's overlap with the known literature is concentrated in ways title-level matching does not see [3, 4]. The funnel then removed 460 of the 495 survivors (SMACT validity 446, stability screen 154, with overlap; one polaron block), leaving 35 (7.1%). Our borrowed stability model disputes the GNoME stable label on 31.1% of the sample; we report that as a disagreement rate between two models and make no claim about which is right. None of the 35 funnel survivors pairs a favorable sustainability profile with a checkable application fit. A uniform sample of the flagship generated library, passed through this stack, yields no candidate this pipeline could defensibly act on, which is a statement about the pipeline and the library jointly, and a base rate any triage system in this field has to beat.

TABLE IV. The pre-registered offset audit. Each couple is computed as a paired vc-relax on the local bench under the anchor-scale settings (Sec. VIII); $\delta = V_{\mathrm{QME}} - V_{\mathrm{exp}}$. The $LiMn_2O_4$ pair is a pre-registered stretch compound: if either half converges outside its registered spin manifold, the verdict is issued on the $n = 3$ core and the exclusion is reported. The verdict gate, fixed before any run: $\mathrm{sd}(\delta) < 0.15\,\mathrm{V}$ means the scale is calibratable and voltages publish with the calibrated offset and error bars; $\mathrm{sd}(\delta) \geq 0.3\,\mathrm{V}$ retires absolute-voltage claims pipeline-wide in favor of ranking-only language; the zone between is an operator decision documented at sync time.

Couple	V_{exp} (V)	Status
$LiCoO_2 \longrightarrow Li_{0.5}CoO_2$	4.0 (3.9–4.1)	done (anchor)
$LiFePO_4 \longrightarrow FePO_4$	3.45 [18]	running[a]
$Li_2FeP_2O_7 \longrightarrow LiFeP_2O_7$	3.5 [19]	running[a]
$LiMn_2O_4 \rightarrow \lambda\text{-}MnO_2$	4.05 [20, 21]	stretch[b]

[a] Queued or running at submission; the verdict is inserted verbatim from the committed analysis output once the six-job set is complete.
[b] Pre-registered stretch compound with a committed exclusion path (Sec. VI).

VI. RESULTS C: CALIBRATING OUR OWN BENCH (PRE-REGISTERED, VERDICT PENDING)

Sections IV and V indict the reference scale we and others train against; consistency requires applying the same standard to our own DFT bench. We have pre-registered, and at the time of writing are executing, an offset audit of the QME PBE+U scale against experimental plateaus on known Li-ion couples. The design is committed and the decision rule is fixed; no result from it appears in this paper.

The audit computes the average voltage of four couples with unimpeachable experimental plateaus at the same redox couples the bench would compute: $LiCoO_2 \longrightarrow Li_{0.5}CoO_2$ (existing anchor, measured window 3.9–4.1 V), $LiFePO_4 \longrightarrow FePO_4$ (3.45 V [18]), $Li_2FeP_2O_7 \longrightarrow LiFeP_2O_7$ (3.5 V [19]), and $LiMn_2O_4 \rightarrow \lambda\text{-}MnO_2$ (4.05 V [20, 21]), the last as a stretch compound whose exclusion path was committed in advance because its true low-temperature ground state is charge-ordered [22] and the ferromagnetic-cubic convention adds a documented model-choice uncertainty (Table IV). Lithium counts per cell are asserted from the staged structures, not formulas, and the committed analysis script refuses to produce a verdict on any mismatch or on an incomplete job set.

At submission time the audit's six relaxations are partway through their queue on the local scheduler, and the pre-registered analysis script has not been run; under the protocol's own rules a partial result is no result. The verdict, whichever way it falls, will be inserted verbatim from the committed analysis output: a calibratable scale with stated error bars, or the retirement of absolute-voltage language from our own ledger. Both outcomes are pre-accepted, and the gate exists precisely so that the answer cannot be negotiated after the numbers arrive.

VII. DISCUSSION

The findings travel beyond this pipeline in proportion to how common its ingredients are, and its ingredients are the field's defaults: a GNN property screen trained where data is plentiful and deployed where it is not, computed reference voltages standing in for measurements, and a novelty notion defined by absence from the databases at hand. On the evidence here, each default fails in a specific, measurable way. The screen's residual is structured, not noisy, so the standard additive recalibration is not merely insufficient but counterproductive: it moves error between voltage regimes. The reference scale is offset from experiment by about half a volt in this chemistry, in the same direction on every row we could decompose, so benchmark MAEs quoted against computed references understate true error for exactly the high-voltage chemistries where screening matters most. And the base rate of already-published compositions in a targeted substitution space is high enough ($\geq$70%) that a screen with perfect accuracy would still mostly re-rank known materials.

We are deliberate about what the decomposition result does not show. It does not show that MP voltages are wrong by 0.54 V in general; $n = 2$, one chemistry, first-charge references. It shows that on the only rows where the comparison was possible at polymorph resolution, the reference term dominated the model term, and that our own independent PBE+U calculation reproduces the offset's direction and rough size on a third compound. The honest generalization is conditional: models trained toward GGA+U intercalation voltages should be presumed to inherit a systematic deficit against experiment in transition-metal phosphate chemistries until a per-chemistry offset audit shows otherwise. That presumption is cheap to test and expensive to ignore.

The novelty result has a constructive reading. Because the prior-art layer is precision-first and downgrade-only, its verdict for any surviving candidate is not "novel" but "no prior art found within the stated coverage": named sources, named match rules, a dated coverage statement, and a hand-measured precision on known and fictional controls. We think this bounded-coverage absence claim is the strongest novelty statement an automated system should ever issue. The alternative, treating absence from a convenient corpus as discovery, is how a field ends up re-announcing 1980s solid-state chemistry, and our own targeted space was 70% re-announcement before the screen ever ran. Subscription structure databases we lack (notably ICSD) would raise coverage, and we state plainly that a zero-budget instantiation cannot close that gap; it can only declare it.

The strongest objection we know how to state is this: the entire exercise is self-referential: a five-anchor ledger of voltage-only results on known compounds, four of them at couples no electrolyte can cycle, gated by a model trained on those same anchors, validated by the same operator who built the system, on six experimental points partly contaminated by first-charge kinetics; nothing here is independently verified, so why should the negative result be trusted more than the system it indicts? Our answer is the protocol, not the pipeline. The thresholds, metric, and failure modes were committed before the metrics existed and are quotable from the repository history; the conservative metric was designed against our own favor; the curation is quote-anchored to fetched sources with a per-row audit trail; an independent human audit of the most verdict-sensitive rows confirmed five of five extractions and *strengthened* the negative verdict by excluding a row that flattered the model; and every number in this paper, including the unflattering ones, resolves to a committed artifact. A skeptic who distrusts our system should distrust it symmetrically: the same self-referential gates it passed (anchor-fit, held-out cross-validation on its own DFT) are the gates we are arguing the field should stop treating as validation. The negative result survives the objection because the objection *is* the result.

Several limitations bound all of the above and are conceded rather than argued. The experimental comparison rests on $n = 6$ ($n = 7$ including the grade-C row; verdict unchanged), the decomposition on $n = 2$, and no bootstrap can manufacture sample size that curation could not. One reference ($NaCrO_2$) is a capacity-weighted scalar we computed from author-stated phase windows; plausible protocol choices move it by 0.2–0.3 V in the direction that worsens, not improves, the model's residual. The anchor ledger mixes runs from two Quantum ESPRESSO installations (a retired cloud host and the local bench) under identical cutoffs, pseudopotentials, and U values, with per-run provenance recorded; the pending offset audit re-computes one pair so that its couple sits entirely on the local scale. The stability and embedding model is borrowed, and its 31.1% disagreement with GNoME labels is unvalidated in both directions. The acquisition stage is unevaluated and unevaluable at $N = 5$. The pipeline has never verified a candidate absent from its corpora, so its discovery half is untested by construction. And the operator-validation step, while documented verbatim in the repository, was performed by the system's author; a third-party replication of the curation from the committed CSV is the cheapest external check this work admits and is explicitly invited.

What survives is narrow but load-bearing: a validation protocol that cannot be renegotiated after the fact, a quote-anchored experimental reference set with full provenance for a chemistry where such sets are scarce, evidence that computed voltage references carry a directional, half-volt-scale error into everything trained on them, and one pipeline that now says "not screening-grade" about itself in public, with the artifacts to back it. We retire the Na screen, we do not retrain it on more computed voltages, and our own bench's absolute-voltage language now hangs on a pre-registered audit it may fail. That is what we believe validation discipline looks like when the answer is no.

VIII. METHODS

A. DFT verification bench

All verification runs are spin-polarized PBE+U variable-cell relaxations in Quantum ESPRESSO [12, 13], with wavefunction and charge-density cutoffs of 50 Ry and 200 Ry, Gaussian smearing of 0.01 Ry, local Thomas–Fermi mixing, and Monkhorst–Pack k-grids [23] set per cell by the input stager and recorded with each run (the anchor cells used $3 \times 3 \times 3$; the audit cells use denser grids, e.g. $6 \times 5 \times 3$ for olivine $LiFePO_4$). Hubbard corrections use the Dudarev formulation [24]: on the current QE 7.3.1 bench through the `HUBBARD {ortho-atomic}` card, and on the retired QE 6.4.x cloud host through the legacy `lda_plus_u` interface; U_{eff} values are Fe 5.3 eV, Co 3.4 eV, Ni 6.2 eV, Mn 3.9 eV (literature-standard [6, 7]), and V 3.1 eV (an empirical fit, recorded as such per run). No first-principles U exists anywhere in the pipeline: one linear-response attempt [25] failed to converge and was not repeated, and we flag every U provenance in the database rather than claim otherwise. Pseudopotentials are pinned by file name (Appendix D); the backbone sets are GBRV ultrasoft [26] and PSlibrary PAW/USPP [27].

The average intercalation voltage of a couple with n cycling ions follows the standard total-energy construction [5],

$$V = -\frac{E_{\mathrm{charged+ion}} - E_{\mathrm{charged}} - n\,\mu_{\mathrm{ion}}}{n}\,, \tag{1}$$

with energies in Ry converted at 13.605693122994 eV/Ry, $\mu_{\mathrm{Li}} = -14.4725646547$ Ry from a BCC Li reference run and $\mu_{\mathrm{Na}} = -95.34612073$ Ry from the analogous Na reference, both stored in the run database. Ion counts n are counted from the relaxed cells, never inferred from formulas, and are hard-asserted by the analysis scripts. The parsed energy is the BFGS `Final enthalpy`; the post-convergence single-point rewrite is never used. Spin assignment is pre-registered per run in the ferromagnetic convention standard for GGA+U voltage work [6], with the expected high-spin manifolds and total magnetizations fixed in advance; a pair whose halves converge in different manifolds is excluded rather than mixed, and one ledger row ($Na_3Fe_2(PO_4)_3$, desodiated) converged only under a fixed total magnetization, which is recorded with the run. Calculations executed sequentially on a Mac Mini M4 (local bench) and, for the early anchor runs, on a retired cloud host; the backend of every run is a database field, and identical cutoffs, pseudopotentials, and U values were used on both.

B. Screen, training data, and gates

The voltage screen is a Siamese multi-head GNN over (charged, discharged) crystal-graph pairs with Monte-Carlo dropout for predictive uncertainty. Its training corpus holds 2814 graphs assembled from MP battery pairs, of which 20 pairs are Na chemistries. Candidate relaxation during screening uses M3GNet [28]; stability estimates and the 128-dimensional embedding space come from MACE-MP-0 [10, 11] with property heads. The deployed model is the first PBE+U fine-tune of the v2.4 baseline, trained against the five anchor voltages with elevated anchor weights; activation was gated on the fitted anchor error (0.084 V mean absolute) and on a held-out leave-one-out cross-validation across the anchors. We report the cross-validation gate qualitatively because its fold-level outputs were not committed to the repository, and this paper does not quote numbers without committed artifacts. Both gates measure agreement with the pipeline's own DFT scale.

C. Validation statistics

The Sec. IV A analysis holds out 74 MP battery entries whose discharged identifiers are absent from the training corpus, with family-restricted leave-one-family-out bias correction over families with at least ten members and a 10,000-resample bootstrap [29] (seed 42) for confidence intervals. The Sec. IV B analysis uses leave-one-out additive bias correction, as pre-registered for $5 \leq n < 10$, with a 10,000-resample bootstrap (seed 20260609); all seven predicted rows are held out by construction (none is in the training corpus, verified by composition mapping rather than identifier string match after the collision of Sec. V B). The primary metric, fixed before computation, is the upper edge of the 95% bootstrap interval on the bias-corrected held-out MAE.

D. Curation and audit

Literature values were extracted only from sources fetched during the curation run, with the supporting sentence or arithmetic recorded per row, the reference electrode checked to be Na metal, and the polymorph lattice-verified against the ingested MP structure. Evidence grades A–C are defined in Sec. III; grade D rows were dropped. The operator audit verified the five most verdict-sensitive rows against primary sources (five confirmed) and excluded the maricite row for phase identity; the audit record, including one clerical heading error, is carried verbatim in the repository and summarized in Appendix B.

E. Prior-art layer and base-rate probes

The layer searches OpenAlex and Crossref at title level with composition-verified matching (exact, deintercalated framework, or element set), an offline structural-family screen, and exact reduced-composition lookup against the Crystallography Open Database via OPTIMADE; patent and ICSD sources are declared stubs that fail open. Its precision audit uses 8 known cathodes and 5 fictional compositions. The GNoME probe draws a uniform random sample ($N = 500$, seed 42, identifier list pinned in the repository) from the SHA-pinned public stable-materials summary (554,054 rows), screens it with the unmodified layer, hand-audits every flagged hit, and then applies the staged funnel (SMACT validity [9], borrowed-model stability, $E_f < 0$ and hull distance < 0.15 eV/atom, and a non-blocking polaron screen) to the prior-art-absent remainder, using the GNoME-relaxed structures from the public release. All probe inputs, outputs, and seeds are committed.

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Data availability. All committed artifacts cited in this paper, namely the pre-registration documents, the curated reference set with quote anchors, the per-row results, the analysis scripts, and the figure-generation code, are available at `https://github.com/Krishnatejavepa/qme-paper-validation`[1] at the tagged commit; the ancillary files of this submission mirror the validation set and analysis scripts.

Appendix A: Curated experimental reference set

Table V lists the nine literature-validated rows with polymorph, evidence grade, reference voltage, and DOI. Rows without an MP structure are excluded from metrics. The $NaCrO_2$ scalar is the capacity-weighted average of the author-stated phase windows, $(0.25 \cdot 2.85 + 0.35 \cdot 3.43)/0.60 = 3.19\,\mathrm{V} \rightarrow 3.20\,\mathrm{V}$ [30]; common fixed-window cycling protocols (2.5–3.6 V) yield 2.9–3.0 V, which would increase the row's +0.84 V residual and leave every verdict unchanged. The two $NaCoPO_4$ values are first-charge averages (Sec. V A); the 0.1 V internal spread between the abstract and figure readings of the β polymorph is recorded in the curation file. Primary sources for the rows not already cited in the text: $Na_4Fe_3(PO_4)_2(P_2O_7)$ [31], $Na_3V_2(PO_4)_2F_3$ [32], $Na_2FeP_2O_7$ [33], and alluaudite $Na_2Fe_2(SO_4)_3$ [34].

Appendix B: Curation log: corrections, drops, and the audit record

Machine validation corrected six MP identifiers from the human curation proposal: $NaFePO_4$ mp-755097→mp-19226 (the former is $Fe_6O_5F_7$); $Na_2FeP_2O_7$ mp-19426→no MP entry (the former is $CaWO_4$); $Na_3V_2(PO_4)_2F_3$ mp-755519→mp-694937 (the former is $Li_3MnCr_3O_8$); $NaCrO_2$ mp-19427→mp-578604 (the former is $Na_7(CoO_3)_2$); Na_2FePO_4F mp-22162→mp-1194940 (the former does not resolve); and the in-training flag for $Na_4Fe_3(PO_4)_2(P_2O_7)$ was a battery-identifier namespace collision with a Cr_3O_8 pair. Five compounds were dropped rather than estimated: maricite $NaMnPO_4$ (no defensible average voltage exists; the best-enabled material reaches ∼30% of theoretical capacity with 2.4 V hysteresis [35], and maricite inactivity is independently reported [16]); $NaVPO_4F$ (full-cell-versus-hard-carbon voltage, disputed phase identity, and an MP entry matching neither claimed phase [17]); $NaFeSO_4F$ (synthesis and structure literature only [36]); $NaNi_{0.5}Mn_{0.5}O_2$ (window stated without an average and no ingestible MP structure [37]); and P2-$Na_{2/3}MnO_2$ (no matching MP structure; structural literature only). The proposal's Na_2FePO_4F value also moved from the Li-cell number 3.5 V [38] to the Na-cell plateaus of Kawabe *et al.* [39], averaged to 2.985 V.

The operator validation record is carried verbatim in the repository file `STEP3B_OPERATOR_VALIDATION_2026-06-09.md`. Two of its features are preserved rather than edited, per the carried-record rule: its Section 2 heading reads "$Na_2FeP_2O_7$" over content that describes the Na_2FePO_4F row (a clerical error flagged in the file's machine note), and its Section 3 illustrates the post-hoc nature of any $NaCrO_2$ scalar with the midpoint arithmetic $(3.1 + 3.7)/2 = 3.4\,\mathrm{V}$; the canonical value used throughout this paper is the capacity-weighted 3.20 V of Appendix A, and the illustration's existence is part of why that row carries grade B rather than A.

In the earlier Step-3 attempt on the 74-compound held-out set, 0 of the 26 polyanionic phosphates had a citable experimental average voltage (the F5 stop condition fired at 100%, against a 50% threshold): 18 had no Na-cell publication of any kind, 3 had structural crystallography only, 2 matched only related-but-

[1] [OPERATOR: confirm the repository is public before submission.]

TABLE V. The curated Na-ion cathode reference set. Grade: A = author-stated average in fetched text; B = one arithmetic step from author-stated plateaus or windows (calculation recorded); C = figure read-off (low confidence). The maricite row is excluded from canonical metrics by the operator audit (phase identity: the cycled phase is amorphous). Rows marked "no MP entry" are literature-validated but cannot be predicted by the structure-ingesting screen.

Compound	Polymorph	MP id	Family	Grade	DOI	$V_{\rm lit}$ (V)
$NaFePO_4$ [a]	maricite	mp-19226	phosphate	C	10.1039/c4ee03215b	2.60
$Na_4Fe_3(PO_4)_2(P_2O_7)$	$Pn2_1a$	mp-1203835	phosphate	A	10.1021/ja3038646	3.20
$NaCoPO_4$	ABW $P2_1/n$	mp-562796	phosphate	A	10.1016/j.jssc.2020.121766	4.50
$NaCoPO_4$	β $P6_5$	mp-683773	phosphate	A	10.1016/j.jssc.2020.121766	4.30
Na_2FePO_4F	Pbcn	mp-1194940	fluorophosphate	B	10.1016/j.elecom.2011.08.038	2.985
$Na_3V_2(PO_4)_2F_3$	NVPF	mp-694937	fluorophosphate	B	10.1038/s41467-019-08359-y	3.95
$NaCrO_2$	O3 $R\bar{3}m$	mp-578604	layered oxide	B	10.1021/acs.chemmater.5b04626	3.20
$Na_2FeP_2O_7$ [b]	$P\bar{1}$	n/a	phosphate	A	10.1016/j.elecom.2012.08.028	3.00
$Na_2Fe_2(SO_4)_3$ [b]	alluaudite	n/a	sulfate	A	10.1038/ncomms5358	3.80

[a] Excluded from canonical metrics by the operator audit (phase identity; Sec. IV B).
[b] No MP structure exists; literature-validated but not predictable by the screen.

different compounds, and 3 (the $NaCoPO_4$ polymorphs and maricite $NaMnPO_4$) had cathode literature without an extractable average. The per-compound search log is the committed file `step3_curation_log.csv`.

Appendix C: Pre-registered definitions, verbatim

From the Step-2 pre-registration (committed at `bd254c2` before any bias-corrected metric was computed). Verdict tiers: screening-grade, held-out MAE < 0.30 V; ranking-only, 0.30–0.50 V; not screening-grade, > 0.50 V. Primary metric: "held-out MAE on the bias-correction-validation set ...reported as a conservative lower bound, specifically the upper edge of the 95% bootstrap confidence interval." F1: "If per-family bias estimates differ by more than 0.15 V ...a single global additive bias correction does not generalize ...a clean tier verdict is not issued." F2: in-sample contamination of the bias estimate invalidates the verdict. F3: "Held-out set has fewer than 20 compounds ...mark the verdict as PROVISIONAL." F4: mixed bias signs across families block any tier. Revision triggers: an in-box experimentally measured average voltage disagreeing with the corrected prediction by more than 0.4 V downgrades the verdict one tier; "two such cases retire the bias-corrected predictor entirely."

From the Step-3 protocol (fixed before curation began; committed with the Step-3 report at `1f5325c`): F5: "if >50% of polyanionic_phosphate compounds ...cannot be cited from primary literature, the subset is not defensibly 'experimental.' STOP and report; do not invent or estimate citations to hit the count."

From the offset-audit pre-registration (committed at `3119728`, before any audit job ran): "PASS: sd(δ) $<$ 0.15 V across the offset points $\rightarrow$ the scale is calibratable per chemistry family; publish voltages with the calibrated offset and error bars. FAIL: sd(δ) $\geq$ 0.3 V $\rightarrow$ absolute-voltage claims are retired everywhere (ranking-only language) ...GRAY ZONE: 0.15 $\leq$ sd $<$ 0.3 V $\rightarrow$ operator decision, documented at sync time." The $LiMn_2O_4$ stretch exclusion and the off-manifold spin exclusion rule are part of the same committed document.

TABLE VI. The complete verified ledger at submission. Every row is voltage-only (kinetics and hull screens pending); anchor weight 0 marks rows that can never enter a retrain. The $Na_3Fe_2(PO_4)_3$ row is computed-only: the calculated couple is Fe^{3+}/Fe^{4+} desodiation, for which no experimental average voltage exists at curation grade A or B (the measured ~2.5 V plateau belongs to the Fe^{3+}/Fe^{2+} insertion couple [40]), so it anchors nothing and touches no experimental claim.

Compound	V (V)	Role	U prov.	Backend
$LiCoO_2$	4.12	anchor	lit.	cloud
$LiFe(PO_3)_4$	5.687	anchor	lit.	cloud
$LiNiP_2O_7$	5.232	anchor	lit.	mixed
$LiFeP_2O_7$	5.275	anchor	lit.	mixed
$Li_2Fe(PO_3)_5$	5.6115	anchor	lit.	mixed
$Na_3V_2(PO_4)_3$	2.8957	touchpoint	emp.	mixed
$Na_3Fe_2(PO_4)_3$	3.5976	computed-only	lit.	local

Appendix D: Ledger, Hubbard parameters, and pseudopotentials

Hubbard $U_{\rm eff}$ (eV): Fe 5.3, Co 3.4, Ni 6.2, Mn 3.9 (literature-standard); V 3.1 (empirical fit; flagged per run). Pinned pseudopotential files for every element in this work: Li `li_pbe_v1.4.uspp.F.UPF`, Na `na_pbe_v1.5.uspp.F.UPF`, Fe `Fe.pbe-spn-kjpaw_psl.0.2.1.UPF`, Co `Co_pbe_v1.2.uspp.F.UPF`, Ni `ni_pbe_v1.4.uspp.F.UPF`, Mn `mn_pbe_v1.5.uspp.F.UPF`, V `v_pbe_v1.4.uspp.F.UPF`, P `P.pbe-n-rrkjus_psl.1.0.0.UPF`, O `O.pbe-n-kjpaw_psl.0.1.UPF` [26, 27]. Registered spin manifolds for the offset audit: $LiFePO_4$ $4\times Fe^{2+}$ high-spin, total 16 μ_B; $FePO_4$ $4\times Fe^{3+}$ high-spin, 20 μ_B;

$Li_2FeP_2O_7$ 16 μ_B; $LiFeP_2O_7$ 20 μ_B; $LiMn_2O_4$ mixed-valence $2\times Mn^{3+}+2\times Mn^{4+}$, 14 μ_B with Jahn–Teller distortion allowed by free vc-relax; λ-MnO_2 $4\times Mn^{4+}$, 12 μ_B.

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